# Varadirek Sethichaiyen (Direk)

**AI Application Engineer | Full-Stack TypeScript, Node.js & React**
sethidirek@gmail.com | +66 94 646 5634 | English / Thai

## SUMMARY

Full-stack AI application engineer with 4+ years building and operating production web applications, APIs, and responsive user interfaces. Strong in TypeScript, Node.js, and React, with hands-on delivery of LLM integrations, RAG, embeddings, vector search, multi-step tool calling, prompt systems, structured outputs, and streaming conversational UI. Experienced owning secure systems end-to-end across sensitive identity, health, estate, and payment data, cloud integrations, observability, incident response, and continuous improvement after go-live.

## TECHNICAL SKILLS

**Primary:** TypeScript, Node.js, React 19, Express.js, Fastify, HTML/CSS, Tailwind CSS

**GenAI Applications:** OpenAI APIs, Vercel AI SDK, RAG, embeddings, MongoDB Atlas Vector Search, document chunking, agents, tool/function calling, prompt templates, structured output, streaming chat/UI

**Backend & Integration:** REST APIs, PostgreSQL, MongoDB, Mongoose, Convex, Redis queues, AWS SQS, webhooks, scheduled jobs, batch workflows, JWT, OAuth, RBAC

**Frontend Systems:** TanStack Router/Query/Store, React Router, Radix UI, shadcn/ui, responsive and accessible UI, component architecture, dashboards, forms, data tables

**Security & AI Safety:** Zod validation, rate limiting, content filtering, output validation, sensitive-field redaction, safe model-output rendering, audit logging, fallback behavior

**Cloud, Quality & Observability:** AWS, Railway, Cloudflare, Vercel, GitHub Actions, CI/CD, Docker fundamentals, Vitest, Sentry, structured logging, metrics, tracing, token/latency/cost telemetry

## COLLABORATION & COMMUNICATION

- Owns production modules from UX feasibility and API design through implementation, deployment, incident troubleshooting, and stabilization after launch.
- Works closely with product and content teams to turn AI capabilities into reliable user workflows; comfortable with Agile delivery, code review, and explaining technical tradeoffs.
- Uses AI-assisted engineering to accelerate implementation and automated tests while personally validating behavior and owning production outcomes; also teaches AI workshops for children with disabilities.

## EXPERIENCE

### Evaheld | Full-Stack Product Engineer | Australia | evaheld.com

**February 2025 - Present**

- Build and operate **Evaheld Legacy Vault**, an AI-guided platform serving about 3,000 visitors per day across personal legacy, health and care, estate planning, identity documents, and trusted-party workflows.
- Developed Charli using the Vercel AI SDK and multiple LLM providers, with streamed responses, session context, prompt templates and versioning, structured outputs, abort handling, fallback behavior, and responsive React chat experiences.
- Implemented RAG by normalizing and chunking user documents, generating batched OpenAI embeddings, and building tenant-scoped MongoDB Atlas vector search across vault content and conversation history.
- Built multi-step AI tools for content, rooms, sharing, contacts, trusted parties, checklists, requests, notifications, and emergency cards, with Zod-validated calls and interactive model-driven UI.
- Delivered guardrails and observability including safe Markdown and structured-UI rendering, content and output validation, rate limits, sensitive audit-field redaction, Sentry, structured logs, tool-failure tracing, and admin dashboards for prompt, token, latency, and cost telemetry.

### Evaheld Memorials | Full-Stack Product Engineer | Australia | evaheldmemorials.com

**2026 - Present**

- Built a production memorial platform serving about 500 visitors per day across responsive public/private memorials, onboarding, media, collaboration, moderation, books, videos, tributes, and condolences.
- Implemented LLM-powered onboarding and content generation with prompt templates, schema-constrained structured outputs, validation, recovery paths, and editable React interfaces.
- Integrated React, Fastify, MongoDB, Stripe webhooks, Redis queues and scheduled jobs, Resend, AWS-compatible storage, Remotion/Lambda rendering, and production deployment flows.

### Real Estate Platform | Full-Stack Engineer | Australia

**May 2026 - Present**

- Building responsive, component-based listing and management workflows with TanStack, React, TypeScript, Tailwind CSS, Cloudflare, and Railway.

### Freelance Full-Stack Developer | Master Eater, Avenure, MUIC LMS

**January 2022 - July 2024**

- Delivered React and Next.js products from Figma wireframes through responsive production UI, backend logic, authentication, content management, deployment, and support.
- Built a university learning management system and commercial websites with dashboards, forms, editable data tables, filtering, sorting, and accessible mobile layouts.

## SELECTED PERSONAL PRODUCT BUILDS

**Bothlines.com** - Embeddable AI customer-chat SaaS with real-time conversations, teams, invitations, RBAC, support tickets, audit logs, Stripe, Resend, PostgreSQL, Drizzle, TanStack Start, and shadcn/ui.

**PasteWords.com** - File-sharing SaaS built with React, PocketBase, Google AdSense, VPS/Coolify deployment, and focused file-sharing UX.

## EDUCATION

**University of Technology Sydney** & **Mahidol University International College** - Bachelor of Science in Computer Science, 2025

## REFERENCES

References available upon request.